

A. Artifact and Protocol Ledger

This appendix records the evidence used in the main text. The pilot ran from exact source commit 50d09184021074e10985eee4c85951ab25e3ba9c; the research specification SHA-256 is 9dea3391d6285a2ab7590a3cf18ad0f76992585d31d047eca892e63fb97a92d3. Dataset content SHA-256 is bc9b78ca5e3b95f91bf34bbc3641a3d6e3f92338b4e3d97065165157848cfc48; split semantic SHA-256 is 3f909d7aaf10ee959aaebf4538e82fb4f391cba4cad5f5fd44850939629ef457; and the foundation checkpoint SHA-256 is 2ddb079cee0e131eaa6912ba581b394551ead283e95c99cfe78c605d10b5736.

Table 2. Primary local evidence files and current SHA-256 values. The run stopped at the original D0 gate, so these numerical artifacts are recorded in a separate paper ledger rather than misrepresented as a completed campaign manifest.

Evidence	SHA-256
Zero-shot validation metrics	8a775d771ac64ea8a95ecd51ad999cd29543a67d27e8b679e16d4b512c014ad3
Adapted validation metrics	ba67f9216c3cacfc2036d9d42a15fd76f6659a0adb619eab0ca46595d524d427
L4 numerical calibration	27179f9a90000064a7bfb25ed707f7faa411f1448db319c8c14c03ad0c5e7d35
D0 pool scores	52c9a97592575c5501d579a7604e011dd8513bbc8b34d4221d2b6b7900dff18e
Failed original gate	60fc5938e62c60ca8da953388820a90b594c317a3529232f63b60889ccf45221

B. Metric Definitions

Let $\Delta \mathbf{F}_{xi} = \hat{\mathbf{F}}_{xi} - \mathbf{F}_{xi}$ for atom i in configuration x . Force-component MAE and RMSE aggregate absolute and squared Cartesian component errors over all atoms. Vector P95 is `numpy.quantile` with $q = 0.95$ and method `linear` over $\|\Delta \mathbf{F}_{xi}\|_2$. Energy MAE averages $|\hat{E}_x - E_x|/N_x$ over configurations. The softening slope is

$$\beta = \frac{\sum_x w_x \sum_i \mathbf{F}_{xi} \cdot \hat{\mathbf{F}}_{xi}}{\sum_x w_x \sum_i \|\mathbf{F}_{xi}\|_2^2}, \quad w_x = (3N_x)^{-1}, \quad (2)$$

so every configuration has equal total component weight. The magnitude-bias diagnostic is the median of $\|\hat{\mathbf{F}}_{xi}\|_2 - \|\mathbf{F}_{xi}\|_2$.

C. Per-Seed Results and Runtime

The corresponding final model SHA-256 values are 6b95301deee9abac63b0cebefa837af7e6847485176a

Table 3. Final validation metrics for each adapted model. Runtime and memory are manifest-reported training values.

Seed	Force MAE	Vector P95	β	Time (s)	MB
42	0.13566	0.47390	0.62426	44.99	2005.65
43	0.13847	0.48363	0.61639	40.06	2005.66
44	0.13333	0.46595	0.63077	40.08	2005.66

5def99910446e431140c, 0f530b5c7ac59be3146df096df2ba10262f9a1b970f32abaf0ce1eb724183674, and d4d27d68918740598c02f89b75ac969ef85617c2b906315ebc7fee313ebb333a.

D. Explicit Non-Claims

This manuscript does not report an acquisition round, selected-set label efficiency, a frozen-test result, a stress-test result, a VESSL replication, statistical significance, or acceptance/rejection of the tail-risk hypothesis. The recorded top-36 gate is a post-failure, pre-label protocol amendment. Zero pool labels were revealed before the amendment, and the research question, arms, budgets, training settings, endpoints, and protected-test policy were preserved.